MED Analysis

In this section we report one experiment. The question is simple. The polarity tagger gives a polarity to every word of a sentence. How good is this information for monotonicity inference? To answer this, we used the MED dataset [2]. MED is a well-known dataset for monotonicity inference. We took the pairs of MED that our lexicon can handle. These are 787 premise–hypothesis pairs. For these pairs, we polarized the 447 premise sentences that occur in them. Then we checked whether the polarities support the relation between premise and hypothesis. The gold labels of MED are our reference.

The MED dataset

MED, the Monotonicity Entailment Dataset, was built by Yanaka et al. [2] to test whether neural models can do monotonicity reasoning. Every item is a premise–hypothesis pair with a gold label, *entailment* or *non-entailment*. Every item also carries a tag that says which kind of monotonicity the pair tests: upward, downward, or non-monotone. Some items carry a second tag for the phenomenon involved, such as lexical knowledge, conjunction, disjunction, conditionals, or negative polarity items.

The dataset has two parts. The first part was collected by crowdsourcing. The authors took declarative sentences from the Parallel Meaning Bank, parsed them with a CCG parser, and computed polarities with the ccg2mono system of Hu and Moss [1]. One polarized phrase of each sentence was underlined. Crowd workers were asked to rewrite the underlined phrase so that it becomes more specific, in three different ways: by adding a modifier, by adding a conjunction, or by replacing a word with a hyponym. Because the rewritten phrase is always more specific, the gold label of each new pair followed from the position alone. In a downward position, the original sentence entails the rewritten one, so the label is *entailment*. In an upward position, it does not, so the label is *non-entailment*. A second crowd task checked these labels. Only the pairs on which all workers agreed with the calculus were kept. Where possible, premise and hypothesis were then swapped, with the label reversed, to balance the two labels. After the swap, the hypothesis is the more general sentence, so an upward position gives *entailment* and a downward position gives *non-entailment*. The second part was collected by hand from eleven linguistics publications, from the GLUE diagnostic set, and from FraCaS. The crowdsourced part has 4,068 pairs and the linguistics part 1,314, so MED has 5,382 pairs in all; of these, 1,820 are tagged upward, 3,270 downward, and 292 non-monotone. In the released file, the two parts are tagged *crowd* and *paper*, and *non-entailment* is written *neutral*.

Setup

From MED we took every pair that the tagger can run with our lexicon. This gave 787 pairs, 510 from the crowdsourced part and 277 from the linguistics part. They come from 447 distinct premise sentences; many premises have several hypotheses. We ran the tagger on each of the 447 premises and recorded the polarity of every word. The hypotheses were not tagged. We then compared the output with MED from two angles.

1. **Direction.** Each MED pair is tagged upward, downward, or non-monotone. In the premise, we located the word or phrase that is replaced in the hypothesis. The tagger gives this

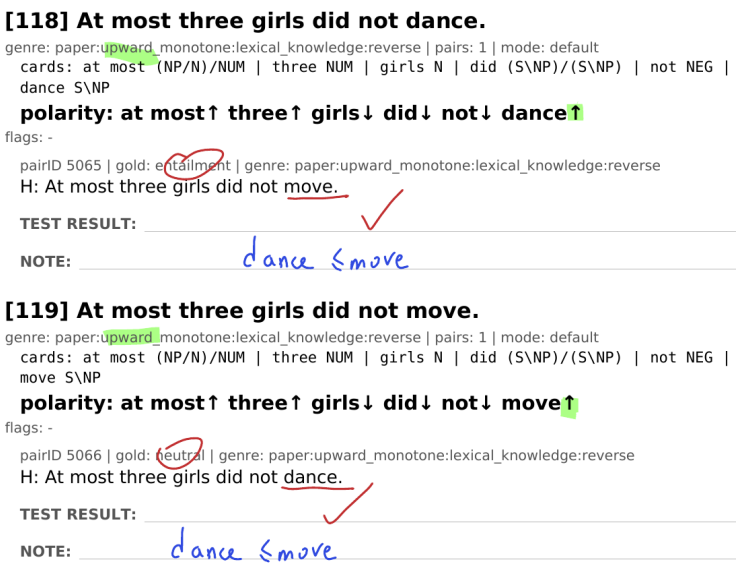


Figure 1: Cases 118 and 119 of the test file (pairs 5065 and 5066), with the manual marks. In both, the premise is tagged *at most*$^{\uparrow}$ *three*$^{\uparrow}$ *girls*$^{\downarrow}$ *did*$^{\downarrow}$ *not*$^{\downarrow}$ *dance*$^{\uparrow}$. In case 118 the verb goes up, from *dance* to *move*, in an $\uparrow$ position, so the gold label *entailment* follows. In case 119 it goes down, so the gold label *neutral* follows.

position the symbol $\uparrow$, $\downarrow$, or $=$. We compared the two. This tells us how far the monotonicity that MED tests agrees with our polarity.

2. **Label.** Each MED pair has a gold label, *entailment* or *neutral*. We checked whether this label follows from our polarity at the replaced position. For example, if the position carries $\uparrow$ and the hypothesis replaces the phrase with a larger one, we say *entailment*. If the replacement goes against the arrow, or the position carries $=$, we say *neutral*. We then compared our label with the gold label.

The browser tool of Section 3 of the full paper carries a reduced lexicon. For this experiment we built a larger version, which covers the words of the MED premises. It is a separate tool in the repository. It has the same thirty-six classes, but each class has more words. The repository also contains the annotated test set and a CSV report built from it. A reader can run the premise sentences in the full tagger and reproduce the result. For every premise, the test file lists the MED tag, the lexical categories, the polarity of every word, and then every hypothesis with its pair ID and gold label. Below, a case number refers to a premise of this file, numbered from 1 to 447, and a pair ID is the ID of the pair in MED. Figure 1 shows two entries, cases 118 and 119, with our marks.

We, the authors, made the comparison by hand, reading each pair. A decision of this kind needs a knowledge base: one has to know, for example, that *dance* is below *move*. So we cannot claim that the check is fully objective. Fixing a convention for the knowledge base is a separate task.

Results

Of the 787 pairs, 712 agree with MED. The remaining 75 pairs lie in 43 of the 447 cases. We go through the disagreeing pairs one by one in the section *The disagreeing pairs* below. Before that, we point out two decisions that we made on the agreeing side. We believe they can be defended, but they are open to discussion, and they may look like deliberate choices.

Decision 1: comparison at the phrase level. In some pairs only one word seems to change. But the word sits inside a phrase that cannot be split, and the replacement really changes the whole phrase. In these pairs we compared the polarity of the phrase, not of the word. This happens with prepositional phrases, verb phrases, and noun phrases. For example:

- PP. Case 25, pair 5081: *Betty lives in Berlin* ⇒ *Betty lives in Germany* (gold: entailment). The tagger gives *in*$^{\uparrow}$ *Berlin*$^{=}$. The name itself cannot be replaced. But the phrase *in Berlin* is below *in Germany*, and the prepositional phrase sits in an ↑ position, so the entailment follows.
- VP. Case 212, pair 1309: *Alessia can't speak English* ⇒ *Alessia can't speak fluent English* (gold: entailment). The tagger gives *speak*$^{\downarrow}$ *English*$^{=}$. The verb phrase *speak fluent English* is below *speak English*, and the verb phrase sits in a ↓ position, so the entailment follows.
- NP. Case 398, pair 283: *Juan and his friends can't help me* ⇒ *Juan can't help me* (gold: entailment). The tagger gives *Juan*$^{=}$ *and*$^{\uparrow}$ *his*$^{\uparrow}$ *friends*$^{=}$. The noun phrase *Juan and his friends* is below *Juan*, and the noun phrase sits in an ↑ position, so the entailment follows.

These polarities are not chosen by hand. In the enriched tree, every phrase carries its own polarity, not only the words. So the polarity of the phrase is read off the tree.

Decision 2: N and NP. Many CCG parsers have a rule that turns the category N into NP. Under this rule, *book* : N and *some book* : NP can be compared under NP category. MED contains many examples of this. Sometimes one case tests *sci-fi novel*, *some sci-fi novel*, *some book*, and *book* together. For example, case 70 has the premise *Every alien read some books* and the two hypotheses *Every alien read sci-fi novels* and *Every alien read some sci-fi novels* (pairs 4304 and 4688). A knowledge base can naturally accept *sci-fi novel* ≤ *book* and *some sci-fi novel* ≤ *some book*. The other two pairs are not so clear. We worked under the following assumption: *sci-fi novel* ≤ *some book* and *some sci-fi novel* ≤ *book*, because the first expression names a more specific set than the second, at least in everyday language. The two sides are not of the same category, and the parser rule N → NP is not a sound rule. A technical study of this point needs separate work. Here we only used this decision to compare MED with the tagger.

The disagreeing pairs

The 75 problematic pairs, in 43 cases, fall into the eight groups below, each given with its number of pairs and cases. Two cases have pairs in two groups and are counted once in the total: case 213 in groups 2 and 6, and case 283 in groups 1 and 4.

1. **Presupposition with *that* and *the*.** Expressions with *that* or *the* sometimes carry a presupposition. In everyday language there seems to be an entailment, but there is no full

entailment. Our lexicon has the design choice that *that* is non-monotone on its N argument. This fits: *that vague explanation* does not become *that explanation*. MED, however, seems to apply the presupposition everywhere. For example, case 282, pair 3619: *I can't accept that explanation* ⇒ *I can't accept that vague explanation* (gold: entailment). In fact, our design of the determiner types agrees with the paper that is the main source of MED (See Table 1 in [2]). So there is no real disagreement. But the entailments in the data cannot be verified from these polarities. There are 29 such pairs, in 10 cases.

2. ***Or* under negation.** $\neg P$ does not entail $\neg(P \vee Q)$. But the annotators seem to have applied this the other way round. For example, case 380, pair 2940: *Tom doesn't usually wear a hat* ⇒ *Tom doesn't usually wear a hat or a cap* (gold: entailment). Here one expects *neutral* where MED says *entailment*. There are 6 such pairs, in 6 cases.

3. **VP *and* under negation.** $\neg W$ entails $\neg(W \wedge S)$. The gold label should be *entailment*. For example, case 371, pair 2879: *I will never write again in English* ⇒ *I will never write and speak again in English* (gold: neutral). The crowd read *never write and speak* distributively, as *neither ... nor*. There are 3 such pairs, in 3 cases.

4. **Not a monotonicity argument.** Some pairs are not really monotonicity arguments. One kind is *S and S*: case 245, pair 2406, *The moon has no atmosphere* ⇒ *The moon has no atmosphere, and the gravity force is too low*. Another kind is *S because/since/as S*: case 372, pair 1958, *I've never flown in an airplane* ⇒ *I've never flown in an airplane because I'm afraid*. Some comparisons make no sense, as in cases 105, 149, and 215: case 105, pair 5372, is *No plumber entered the race* ⇒ *Many men entered the race early* (gold: entailment), and case 215, pair 2117, is *Every crime must be punished* ⇒ *Every criminals must be punished* (gold: entailment). There are 11 such pairs, in 9 cases.

5. **Cases 59 and 305, probable typos in the tag.** In case 59, pair 5141, the premise is *At most three elephants are blue*, the hypothesis is *At most three elephants are navy blue*, and the gold label is *entailment*. The tagger gives *blue*$^{\downarrow}$, and *navy blue* is below *blue*, so the label follows. But the MED tag says *upward*. In case 305, pair 1058, the premise is *I had no friends in Boston*, the hypothesis is *I had no friends in Boston or Texas*, and the gold label is *neutral*. The tagger gives *in*$^{\downarrow}$ *Boston*$^{=}$. Here *in Boston or Texas* is a step up in a $\downarrow$ position, so *neutral* follows. Again the MED tag says *upward*. Both are probably typos in the tag.

6. **Proper names.** In 8 cases a proper name is replaced: cases 115, 213, 238, 239, 270, 312, 313, and 415, with 17 pairs. Our design choice treats a proper name as non-monotone, $=$. MED treats it as upward monotone. For example, case 270, pair 446: *Dan didn't use a mobile* ⇒ *Nick didn't use a mobile* (gold: neutral, tag: upward).

7. **MED is clearly wrong.** In 6 pairs the gold label is clearly mistaken: the pairs of cases 310, 316, 330, 332, 341, and 349. All six replace a noun in a $\downarrow$ position with a larger or an unrelated one and still carry the gold label *entailment*. For example, case 330, pair 2466: *Tom can't drive a car* ⇒ *Tom can't drive a motor vehicle* (gold: entailment).

8. **Generic *a*.** The tagger obtains the generic reading of *a* through *every*: the user writes *every* in place of a generic *a*. MED, however, contains the generic *a* as it is. Case 266, pair 744, *A wolf can not be tamed* ⇒ *A wolf or a bear can not be tamed* (gold: neutral), is therefore a result of the missing generic *a* in the tagger.

From this we conclude that the polarity information produced by the tagger, with its default lexicon, gives the correct entailment in most cases. Generic expressions, presuppositions, proper names, and wider coverage each need a lexicon study of their own. In the long run, we claim, the goal is to fix the monotonicity markings of the types of a sufficiently wide class of lexical items. This lexical design is what an ideal algorithm really needs.

References

[1] Hai Hu and Lawrence S. Moss. Polarity computations in flexible categorial grammar. In *Proceedings of the Seventh Joint Conference on Lexical and Computational Semantics (*SEM 2018)*, pages 124–129, New Orleans, LA, 2018.

[2] Hitomi Yanaka, Koji Mineshima, Daisuke Bekki, Kentaro Inui, Satoshi Sekine, Lasha Abzianidze, and Johan Bos. Can neural networks understand monotonicity reasoning? In *Proceedings of the 2019 ACL Workshop BlackboxNLP: Analyzing and Interpreting Neural Networks for NLP*, pages 31–40, Florence, Italy, 2019. Association for Computational Linguistics.